

Interaction-induced d_{xy} pairing in the checkerboard Hubbard model

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We study pairing in the half-filled Hubbard model on the checkerboard lattice by constrained-path quantum Monte Carlo. Separating the interaction-induced vertex from the uncorrelated part of the pair-field correlator, we find that the vertex vanishes identically in the noninteracting limit in all four channels examined, so any pairing tendency is generated by the interaction rather than inherited from the band structure. The d_{xy} vertex is the largest of the four above a crossover anisotropy δ_c and grows monotonically with interaction strength, from 3.347(8) at $U = 2$ to 5.952(22) at $U = 4.5$, peaking at $\delta = 0.4$ in both cases. The crossover is nearly independent of interaction strength and becomes more so with system size, its spread falling from 0.148 to 0.028 between 8×8 and 12×12 lattices while the mean moves from 0.235 to 0.192, which identifies $\delta_c \approx 0.19$ as a property of the lattice geometry. The d_{xy} vertex grows by a factor between 2.18 and 2.38 over the same range of sizes, while the $d_{x^2-y^2}$ vertex remains positive everywhere and never becomes repulsive. The dominant channel lies outside the basis used in earlier finite-temperature work on a related checkerboard Hamiltonian.

I. INTRODUCTION

Altermagnets combine a vanishing net moment with a spin-split band structure, a combination long assumed to require either ferromagnetism or spin-orbit coupling [21, 23–25]. The symmetry classification and the order-parameter description have since been put on a group-theoretic footing [38, 40, 41], and the state has been identified in MnTe [33, 34, 44], CrSb [35], and RuO₂ [26, 36], though the last of these remains contested [32]. Proposals for altermagnetic superconductivity have followed [28–30, 39]. The splitting follows from the crystal symmetry. The two opposite-spin sublattices are connected by a rotation rather than by a translation or an inversion, so the spin degeneracy that a conventional antiferromagnet enforces is absent.

Most microscopic models of this state build the splitting into the Hamiltonian, usually through a spin-dependent hopping. The splitting is then present already at zero interaction, and the question of whether interactions alone can produce it is left open [37]. The Lieb lattice, whose flat band makes the interacting problem sharp [5, 8, 9], has been studied twice for this purpose, and both papers state the gap directly, noting that unbiased studies establishing an altermagnet as the ground state of an interacting Hamiltonian are lacking [42, 43]. Both use methods with a controlled range of validity, unrestricted Hartree-Fock with exact diagonalization on small clusters in the first case and a functional renormalization group treatment of the weak-coupling limit in the second.

The checkerboard lattice offers a route to the same physics without putting it in by hand. It is a square lattice with diagonal bonds on alternating plaquettes, so the two sublattices are exchanged by a fourfold rotation about a plaquette center but by no translation. The hopping is spin independent and the interaction is a uniform on-site Hubbard U . Any spin splitting therefore appears only once magnetic correlations develop, which

is the emergent route rather than the built-in one.

Here we ask what the interaction does to pairing on this lattice, independently of whether long-range magnetic order develops. We report constrained-path quantum Monte Carlo calculations of the pair-field vertex in four channels at half filling on lattices of 8×8 , 10×10 , and 12×12 sites. The vertex is zero at $U = 0$ in every channel, the d_{xy} channel leads above a crossover anisotropy that converges with system size, and the d_{xy} vertex grows with system size at every interaction strength studied.

II. MODEL AND METHOD

We consider the single-band Hubbard model [1, 27],

$$H = \sum_{ij\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (1)$$

with nearest-neighbor hopping $t_{ij} = t_0$ and diagonal hopping $t_1 + t_2$ along $(1, 1)$ and $t_1 - t_2$ along $(1, -1)$ on one sublattice, the two exchanged on the other. Energies are in units of $|t_0|$, and we set $t_0 = -1$ and $t_1 = 0.3$ throughout. The anisotropy is parametrized by δ through $t_2 = -\delta$, so that $\delta = 0$ restores the plain square lattice with uniform second-neighbor hopping and the two sublattices become equivalent. All calculations reported here are at half filling unless stated otherwise.

The hopping in Eq. (1) is spin independent by construction, so $n_\uparrow(\mathbf{k}) = n_\downarrow(\mathbf{k})$ at $U = 0$ for every δ . Figure 1 shows the geometry together with the mean-field polarization it produces.

We use constrained-path quantum Monte Carlo (CPQMC) [10, 11], an auxiliary-field method [2, 3, 6] in which the constraint removes the exponential sign problem [7, 12] at the cost of a bias. Benchmarks of the same method against exact results for the square-lattice Hubbard model are available [4, 13, 17, 18]. We use a Trotter step $\Delta\tau = 0.01$, a projection length $\beta = 32$,

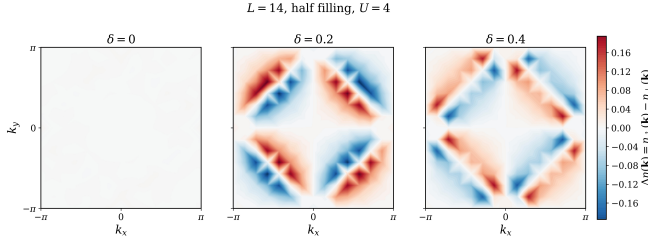


FIG. 1. Checkerboard lattice and the mean-field spin polarization. The nearest-neighbor hopping is t and the diagonal hopping is $t_1 \pm \delta$ on alternating plaquettes, with both spins seeing the same hopping. The momentum distribution difference $\Delta n(\mathbf{k})$ vanishes at $\delta = 0$ and develops a d_{xy} pattern that grows with δ once a moment forms.

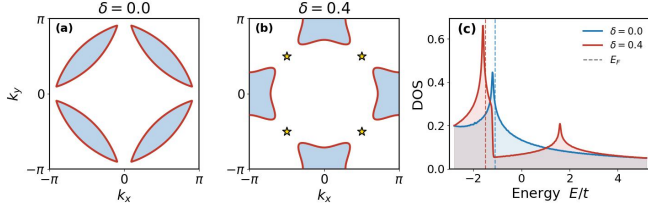


FIG. 2. Fermi surface and density of states of Eq. (1) at $U = 0$. The anisotropy δ moves the van Hove points to incommensurate momenta near the Fermi level, which is the single-particle change that the interacting calculations build on.

and $N_w = 1000$ walkers. The constrained-path approximation introduces a bias controlled by the trial wave function, so we do not describe these results as exact. Appendix A sets out how the trial is chosen, which is not a neutral detail. Doubling the projection length from $\beta = 32$ to $\beta = 48$ changes the order parameter by between 0.1% and 4.1% with no systematic drift, which we take as convergence in β .

III. PAIRING IN THE d_{xy} CHANNEL

We compute the pair-field susceptibility in four channels, on-site s , extended s , $d_{x^2-y^2}$, and d_{xy} , and separate the interaction-induced vertex from the uncorrelated part. The vertex is the meaningful measure of whether the interaction builds pairing, since the full correlator is large and positive already for free electrons. The same separation underlies the long-running question of whether the plain square-lattice Hubbard model superconducts at all [14, 19, 20, 22, 31], and the study of pairing driven by repulsion on other lattice geometries [15, 16]. Figure 3 contrasts the two.

Three features stand out. The vertex is zero in all four channels at $U = 0$, so everything that follows is interaction generated. The on-site s vertex is negative throughout, as a repulsive U requires. The d_{xy} vertex is the largest of the four for $\delta \geq 0.3$ at every U studied, and

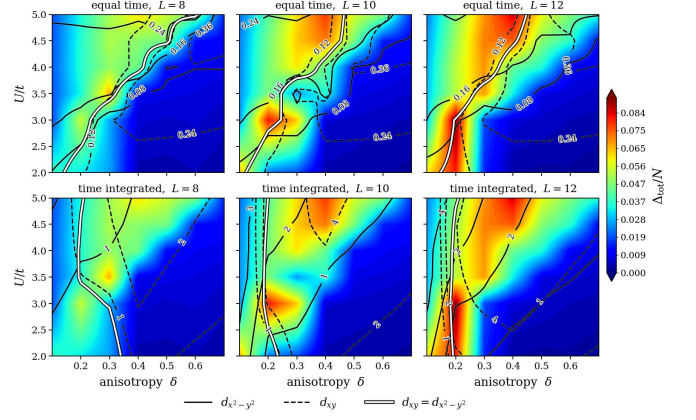


FIG. 3. Equal-time correlator (top row) against the interaction-induced vertex (bottom row) for $L = 8, 10$, and 12 . The channel crossover straightens along the vertex row, which is why the vertex rather than the full correlator is used throughout.

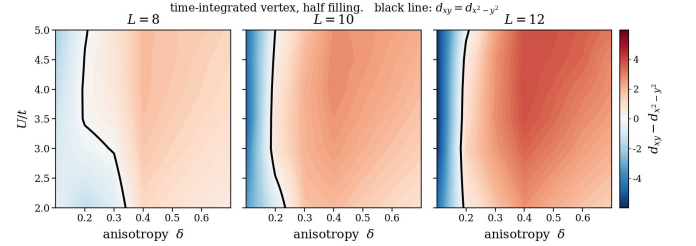


FIG. 4. Difference between the d_{xy} and $d_{x^2-y^2}$ vertices over the (U, δ) plane at $L = 8, 10$, and 12 . The white contour marks where the two are equal. d_{xy} leads everywhere to its right, and the boundary moves very little with size.

it grows monotonically with U , from 3.347(8) at $U = 2$ to 5.952(22) at $U = 4.5$, in both cases peaking at $\delta = 0.4$.

The leading channel changes with anisotropy. For $\delta \leq 0.2$ the extended s channel is largest and for $\delta \geq 0.3$ the d_{xy} channel is. Figure 4 shows the difference between the two d channels directly across three sizes.

IV. SIZE DEPENDENCE OF THE PAIRING VERTEX

The vertex was computed at $L = 8, 10$, and 12 under identical settings, which allows the anisotropy of the channel crossover to be followed with size.

The crossover anisotropy δ_c , defined as the value at which the d_{xy} vertex overtakes the extended s vertex, is nearly independent of U and becomes more so as the lattice grows. Its spread across the six interaction strengths studied is 0.148 at $L = 8$, 0.049 at $L = 10$, and 0.028 at $L = 12$, and the mean value moves from 0.235 to 0.197 to 0.192. A crossover set by the interaction strength would not converge in this way, so we read $\delta_c \approx 0.19$ as a property of the lattice geometry.

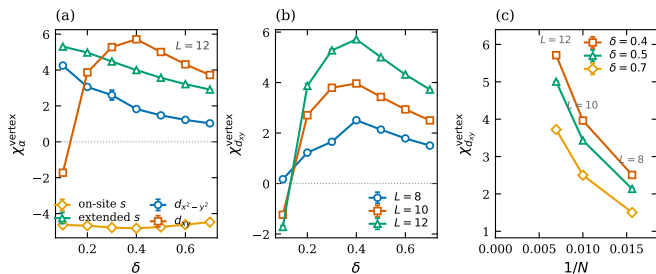


FIG. 5. Interaction-induced pairing vertex at half filling, $U = 4$. (a) The four channels against δ at $L = 12$. (b) The d_{xy} vertex against δ for three sizes. (c) The d_{xy} vertex against $1/N$ at three anisotropies, growing with system size in every case.

The d_{xy} vertex grows with system size at every interaction strength, by a factor between 2.18 and 2.38 from $L = 8$ to $L = 12$. The $d_{x^2-y^2}$ vertex is positive in all 126 cells at all three sizes, so that channel is never repulsive even where it is subdominant. The optimal anisotropy for d_{xy} is $\delta = 0.4$ at essentially every size and interaction strength. Figure 5 collects these three statements.

V. ROBUSTNESS AND COMPARISON

At half filling with periodic boundaries the noninteracting spectrum of Eq. (1) is gapless at every L and δ we examined, so the occupied set is fixed by the trial state rather than by the spectrum. Appendix C reports a repeat with antiperiodic boundaries, which shift the momentum mesh off the high-symmetry points and open a gap. The check is made on the momentum-space spin polarization Δ_{tot} of Ref. [45], which responds to the mesh more sharply than the pairing vertex does. The two boundary conditions agree on the sign of $\partial\Delta_{\text{tot}}/\partial U$ at $\delta = 0.1$ and $\delta = 0.5$ and disagree at $\delta = 0.3$, which we read as the anisotropy where the slope passes through zero.

The closest published work is a determinant quantum Monte Carlo study of the checkerboard Hubbard model that reports d -wave and $d + is$ pairing as the dominant channels [46]. The two Hamiltonians are not the same. That model keeps one diagonal per sublattice and omits the other, whereas Eq. (1) keeps both, and the two coincide only where our second diagonal vanishes, at $\delta = t_1 = 0.3$. We verified that the single-particle spectra agree there to within 4×10^{-15} .

At that point, and at half filling, we evaluated the vertex in the basis of Ref. [46] alongside our own. Within their basis the s -wave channel is the larger of the two at $U = 3, 4$, and 4.5 . Including d_{xy} , which their basis does not contain, gives a d_{xy} vertex larger than either. The comparison is restricted, since Ref. [46] works at density $n = 0.9$ and temperature $T/t = 1/6$ while our calculations are at half filling in the ground state, and under the particle-hole transformation that relates the

two signs of the second-neighbor hopping their density maps to $n = 1.1$. The statement is that the dominant channel we find lies outside the basis used there, not that the two calculations disagree.

VI. CONCLUSION

The interaction-induced pairing vertex of the half-filled checkerboard Hubbard model vanishes at $U = 0$ in all four channels examined and is largest in the d_{xy} channel above a crossover anisotropy $\delta_c \approx 0.19$. The crossover is set by the lattice rather than by the interaction, since its spread across interaction strength falls by a factor of five between 8×8 and 12×12 lattices while its mean barely moves. The d_{xy} vertex grows with system size at every interaction strength, and the $d_{x^2-y^2}$ vertex, while subdominant, is never repulsive.

The dominant channel is one that earlier finite-temperature work on a related checkerboard Hamiltonian did not include in its basis. Whether the vertex develops into a true pairing instability is not settled by the sizes reached here, and the constrained-path bias remains.

Appendix A: Trial wave function and the selection of the constrained path

The constrained-path approximation replaces the sign problem with a bias fixed by the trial wave function, so the trial is part of the method rather than an implementation detail. We take the trial from an unrestricted Hartree-Fock solution of Eq. (1) at the same U , δ , and filling, and we select among the solutions reached from different starting densities by the variational energy

$$E_{\text{HF}} = \text{Tr}(K\rho_{\uparrow}) + \text{Tr}(K\rho_{\downarrow}) + U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle, \quad (\text{A1})$$

with $\rho_{\sigma} = V_{\sigma} V_{\sigma}^{\dagger}$ built from the occupied columns of the mean-field eigenvectors and K the one-body matrix of Eq. (1).

Equation (A1) is not interchangeable with the sum of mean-field eigenvalues. Each eigenvalue already contains the mean field of the opposite spin, so that sum counts the interaction twice and orders competing solutions differently from the energy. The two criteria can select different states, and because the constrained path is fixed by whichever state is chosen, the selection propagates into every measured quantity. At half filling the state chosen by the eigenvalue sum carries the larger staggered moment in every case where the two differ, so the criterion matters for exactly the quantity under study. We report results obtained with Eq. (A1) throughout.

The identity $E_{\text{HF}} = \sum_p \varepsilon_p - U \sum_i \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle$ holds only at exact self consistency and should not be substituted for Eq. (A1). A mean-field iteration that mixes densities between steps does not reach that condition for loosely

converged solutions, which are precisely the ones whose ranking is at issue.

Appendix B: Convergence in projection length

All results use a projection length $\beta = 32$ in units of $1/|t_0|$. Doubling it to $\beta = 48$ at otherwise identical settings changes the order parameter by between 0.1% and 4.1% with no systematic drift in either direction, which we take as convergence.

Appendix C: Boundary conditions

Antiperiodic boundaries shift the momentum mesh off the high-symmetry points and open a gap in the non-interacting spectrum. The sign of $\partial\Delta_{\text{tot}}/\partial U$ agrees between the two choices at $\delta = 0.1$, where both are negative, and at $\delta = 0.5$, where both are positive, with magnitudes matching to within a few percent. At $\delta = 0.3$ the two disagree and the antiperiodic value is consistent with zero. We read $\delta \approx 0.3$ as the anisotropy at which the slope passes through zero, where its sign is most sensitive to the mesh, and we do not quote a slope there.

ACKNOWLEDGMENTS

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